

## NASA making tiny robots to explore other planets

An army of autonomous self-driving robots is being developed by NASA which would be equipped with webcams, Wi-Fi antennae and GPS to explore other planets in the future. The army of robots is codenamed as swarmie. The swarmies function just like an ant colony. When an ant finds out a food source, it notifies rest of the colony through a special signal. Then the other ants jump in and take that food to their nest.

NASA's Kennedy Space Center in Florida has developed the software which controls these swarmies. These robots move out in different directions and start searching for a specific material, such as ice-water on Mars. If any of the rover finds something similar, it uses radio communication to call other robots to help in collecting more samples.

Kurt Leucht, one of the engineers who are associated with the project, said, "For a while people were interested in putting as much smarts and capability as they could on their one robot. Now people are realising you can have much smaller, much simpler robots that can work together and achieve a task. One of them can roll over and die and it's not the end of the mission because the others can still accomplish the task."

The testing phase for these swarmies is just in its primary stage and presently the robots have been programmed just to look for bar-coded slips of paper. In next few months the tests will also include RASSOR, which is a mining robot, designed to dig into alien surfaces and look for valuable materials. These tests will determine how much this software can control the robots when they are functional.

## Intel launches world's smallest 3G modem

Intel has launched the world's smallest 3G standalone modem for the Internet



*The self-driving swarmie robots could be used to search alien surfaces one day (NASA/Dmitri Gerondidakis)*

of Things (IoT). It is likely to find application in home automation, industrial automation and wearable technology.

Intel XMM 6255 is just 300 square millimetre in size, which is as small as one-cent US coin. Intel has packed Intel Power Transceiver Technology in this 3G modem chip. Intel Power Transceiver comprises receiver, amplifier, transmission and power management units in a chip. The company has focused on low power consumption and overheating issues while developing this chip.

Intel claims that XMM 6255 can work even in low signal areas without the need of a stronger antenna. The chip can be used in smartwatches to provide on-board Internet connectivity. For instance, this chip could be used in Gear S smartwatch which comes with 3G SIM card slot. Many other devices can also use this chip for Internet connectivity, just as carbon monoxide monitor and Nest Protect smoke alarm uses Wi-Fi to send warnings to a smartphone. The Wi-Fi can be replaced with the IoT by using the XMM6255 chips.

## After smart watches and eye-wears, here comes a smart shoe

An Indian hi-tech start-up has developed a real-life GPS-enabled smart sports shoe which vibrates to give directions to the wearer. The sneakers can also count the number of steps taken, distance travelled and calories burnt. The smart shoes, called LeChal, are ready to go on sale very soon.

A detachable Bluetooth transceiver is integrated into the shoes, which links to a smartphone app. This app directs the wearer using Google maps and sends a vibrating signal to indicate a left or right turn. Krispian Lawrence, 30, and Anirudh Sharma, 28, who developed the smart shoes founded their start-up Ducere in 2011.

Lawrence says, "We got this idea and realised that it would really help visually challenged people as it would work without any audio or physical distractions. The footwear works instinctively. Imagine, if someone taps your right shoulder,